

# DANIEL LIAN

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Highly motivated graduate student in Computer Science with passion for software engineering, artificial intelligence, machine learning, cybersecurity, and data science. Outside of computer science, I enjoy running and volleyball.

## Education

### University of Michigan

*Master of Science in Engineering in Computer Science & Engineering*

**Aug. 2026 – May 2027**

*Ann Arbor, Michigan*

### University of Michigan

*Bachelor of Science in Engineering in Computer Science - Summa Cum Laude (3.76/4.00)*

**Aug. 2023 – May 2026**

*Ann Arbor, Michigan*

## Experience

### Software Engineering Intern

*Employable AI*

**July 2026 – August 2026**

*McLean, VA*

- Developed 5-step candidate pipeline to streamline job matching process between jobfinders and employers.
- Created Claude-based resume builder and analyzer to create ATS-compliant resumes and provide actionable feedback.

### AI Patents Intern

*United States Patents and Trademark Office*

**June 2026 – August 2026**

*Alexandria, VA*

- Examined artificial intelligence-related patents as part of Technology Center 2100, evaluating novel machine learning architecture and deep learning methods and identifying prior disclosures to generate claim mappings for office actions.
- Developed application to help speed up search process 1000-fold to clear up years-long patents queue backlog.

### Head Instructor

*Fairfax Collegiate*

**June 2023 – August 2025**

*McLean, VA*

- Led 15 STEM classes over 15 weeks, teaching over 100 students from 3rd grade through 10th grade robotics, Scratch programming, advanced math concepts (including middle school math and algebra), and game design.
- Honed communication skills by mentoring junior staff, created friendlier team-based learning environment to adapt to students' diverse learning styles.

## Projects

### AI-Powered Patent Examiner | *Python, RAG, OpenAI*

**June 2026 - Present**

- Designed and developed LLM-based system to speed up searching for prior art and matching patent claims to prior art.
- Achieved 1000x speedup of searching process with minimal token usage, providing minimal-cost solution reducing timescale of search from days to minutes while preserving 0-data retention and complete confidentiality.

### Compas Course Recommender | *TypeScript, HTML, PostgreSQL, Python, Selenium*

**January 2026 - May 2026**

- Designed and developed system using Gemini backend LLM to provide University of Michigan course recommendations and reasoning based on user's degree program, course history, and preferences with iterative feedback mechanism.
- Used PostgreSQL backend to store course data from over 14,000 courses with metrics for interest and workload.

### Cities Chain Discord Bot | *Python, Pandas, MariaDB, SQL*

**July 2023 - Present**

- Designed and implemented a scalable Discord game supporting thousands of players across 1500+ Discord servers using efficient SQL backend data management in MariaDB for seamless gameplay.
- Applied Agile software maintenance and iterative feedback loops to ensure robust gameplay experience.

### Election Prediction Algorithm | *Jupyter Notebook, TensorFlow, Keras, Pandas, NumPy*

**August 2022 - May 2023**

- Implemented Python Tensorflow model in Jupyter Notebook to predict 85% of 2022 United States congressional winners using weighted average of 538 polls, demonstrating robust predictive analytics capabilities.
- Presented findings as part of senior research project to audience of 40 at tjSTAR technology research symposium.

## Technical Skills

**Languages:** Python (Pytorch, TensorFlow, Selenium, Flask, Pandas, NumPy, scikit-learn), SQL, C/C++, HTML/CSS, JavaScript, Java, LaTeX

**Developer Tools/Technologies/Frameworks:** VS Code, Eclipse, Linux, Git, GitHub, MariaDB, Jupyter Notebook

**Certificates:** Amazon Web Services Certified Solutions Architect Associate

## Leadership / Extracurricular

**Extracurriculars:** Asian Intervarsity, WolvSec, Michigan Quizbowl, 3D Printing Club, Volleyball